

Log returns can take on any value from $-\infty$ to $+\infty$ and the distribution of returns is symmetric. However, the percentage returns can only take on values between -100% and $+\infty$. The percentage return data values have a skewed distribution while the log returns distribution is symmetric.

The log returns method is a symmetric measure and provides mathematical simplification in many of our calculations. It is also more intuitive than the percentage change method. For example, if the price for a financial asset changes from 100 to 80 the percentage price change is -20% . However, if the price in the next period changes from 80 to 100 the percentage price change is $+25\%$. So this asset first lost 20% then increased 25% , but it is currently at its original starting value—it is not $+5\%$ overall. But using the log returns method we do not run into this inconsistency. If the stock price changes from 100 to 80 this corresponds to a log change of -22.31% . If the price in the next period increases from 80 to 100, this corresponds to a log change of $+22.31\%$. The sum of these returns states that the current price should be the same as the starting price since $+22.31\% - 22.31\% = 0$ which is exactly what we have.

Price returns have been found to have a log-normal distribution. Thus using the log returns measure is more consistent with the underlying data. Going forward we will use the log returns methodology.

Volatility

Volatility is calculated as the standard deviation of price returns.

$$\sigma_i = \sqrt{\frac{1}{n-1} \sum_{k=1}^n (r_{ik} - \bar{r}_i)^2} \quad (6.6)$$

In the formulation above, there are n historical returns but we divide by $(n-1)$ to ensure an unbiased estimate. This formulation is also called the sample standard deviation.

Covariance

The covariance of returns for two stocks σ_{ij} is a measure of the co-movement of prices.

$$\sigma_{ij} = \frac{1}{n-2} \sum (r_{ik} - \bar{r}_i)(r_{jk} - \bar{r}_j) \quad (6.7)$$

Positive covariance means that the prices will move up and down together and negative covariance means that the prices will move in opposite directions.